

Armazéns de Dados

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Introduction to Data Warehouses

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Summary

- Definition of Data Warehouse
 - Based on four keywords
- Data Warehouse Characteristics
- A Data Warehouse Architecture
- Definition of Data Mart

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Definition of Data Warehouse (DW)

- **Data warehouse is a subject-oriented, integrated, time-variant, non-updatable collection of data used in support of the decision-making process and business intelligence [Inmon and Hackathorn, 1994]**

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Data Warehouse - Subject-Oriented

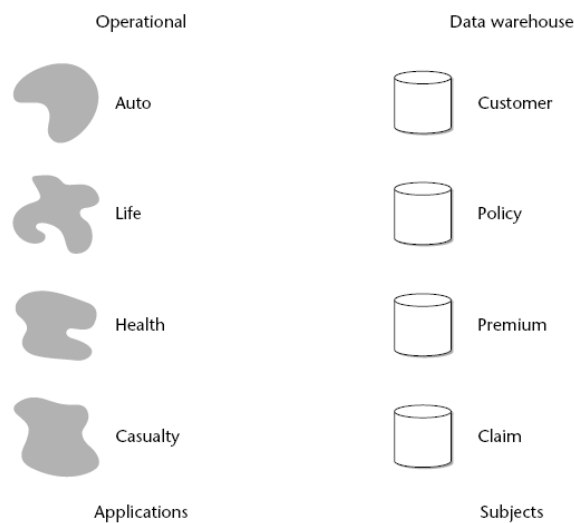
- Organized around key subjects of the enterprise, such as **sales, purchases, inscriptions**
- Focused on the **modeling and analysis of data for decision makers**
 - not on daily operations or transaction processing
- Provide **a simple and concise** view around particular subject issues, **excluding data not useful in the decision support process**

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Data Warehouse - Subject-Oriented

SUBJECT ORIENTATION



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Data Warehouse - Integrated

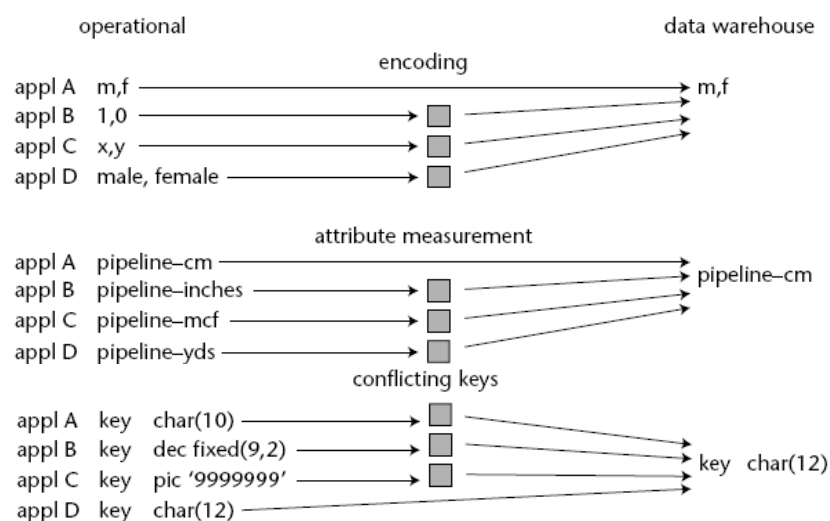
- Constructed by integrating **multiple, heterogeneous data sources**
 - relational databases, text files, binary files, ...
- **Data cleaning** and **data transformation** techniques are applied
 - consistent naming conventions, formats, encoding structures, attribute measures, ... among different data sources (internal and external)
- DW holds the **version of “the truth”**

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Data Warehouse - Integrated

INTEGRATION



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Data Warehouse - Time Variant

- Time horizon is longer than that of On-Line Transaction Processing (OLTP) systems
 - Operational database: current data
 - Data warehouse: provide **information from a historical perspective** (e.g., past 5-10 years)
 - Snapshots of OLTP systems are moved to the DW as a series of data layers – much like **geologic layers**
- Some form of **time marking** to show the moment in time during which **the record is/was accurate**

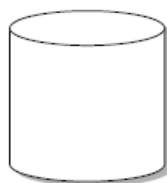
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Data Warehouse - Time Variant

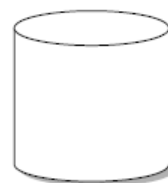
TIME VARIANCY

Operational



- Time horizon – current
- Update of records
- Schema may or may not contain an element of time

Data warehouse



- Time horizon – 5–10 years
- Sophisticated snapshots of data
- Schema contains an element of time

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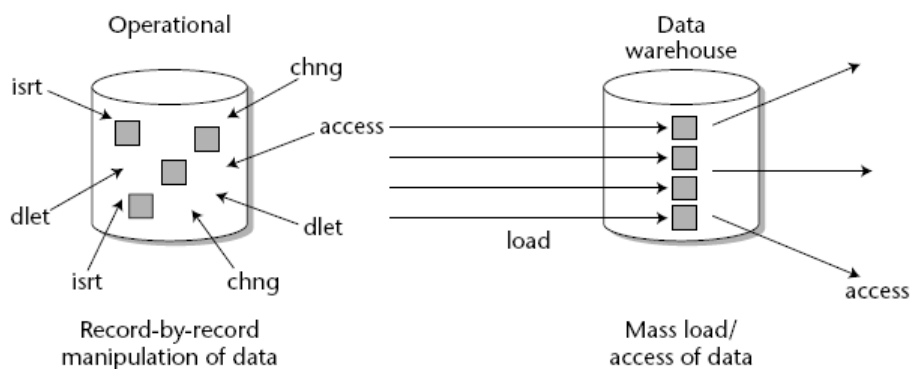
Data Warehouse - Non-Updatable

- **Physically separated store** of data transformed from the operational environment
- **Update of data does not occur** in the DW environment
 - Does not require **recovery** and **concurrency** control mechanisms
 - Requires only two data operations: **load** and **access**
 - **Cannot be updated** by end-users

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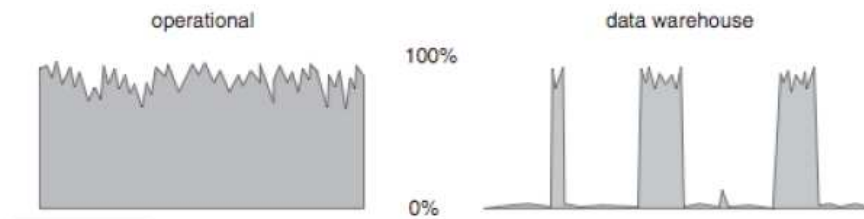
Data Warehouse - Non-Updatable



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Utilization Patterns for OLTP and DSS



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Data Warehouse Characteristics

- **Collection of technologies** for the *knowledge worker* make **better** and **faster** decisions
- Targeted for **decision support**
- Contain **consolidated data**, sometimes from several operational databases, over **long periods of time**
- Tend to be **orders of magnitude larger than operational databases**
- Projected to be **hundreds of gigabytes to terabytes** in size

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Advantages of a Data Warehouse

- Query performance
- Queries not visible outside the data warehouse
- Local processing at sources unaffected
- Can operate when sources unavailable
- Can query data not stored in the OLTP systems
- Extra information at the DW
 - Summarized (aggregated)
 - Historical information
- Consolidated view of organizational data

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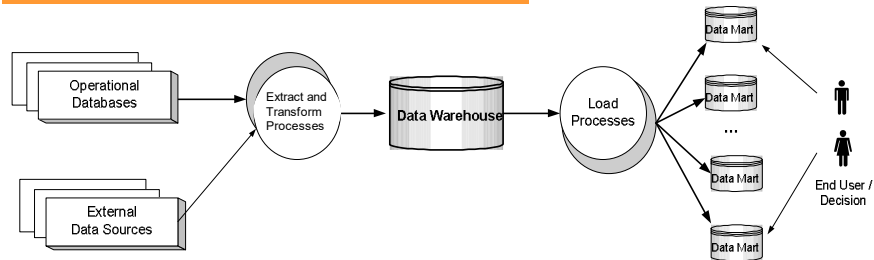
Disadvantages of a Data Warehouse

- Initial cost
- Time consuming development
- Maintenance cost
- Difficulty with heterogeneity of hardware and software

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A Data Warehouse Architecture



Data warehouse is based on a “supply chain” metaphor:

1. “Data raw material” is obtained from “data suppliers”
2. Data is stored in a central “data warehouse”
3. Data is delivered via “data marts” to “data consumers”

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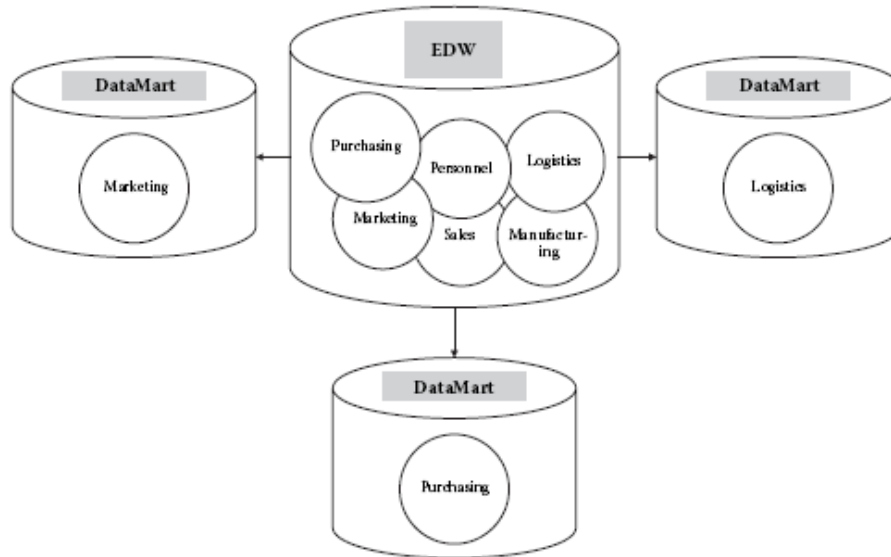
Data Mart

- **Small DW** which contains only a subset of the Enterprise-wide Data Warehouse (EDW)
- DW **limited in scope**
- **Specific to a department or group of users** containing only the data which is relevant
 - Example: **marketing data mart** has only information about customers, products and sales

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Data Mart



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Data Warehouse vs. Data Mart

- Data mart enables fast response to queries
- Data in the data mart is usually more aggregated and less voluminous than in a DW
- Data in a DW is detailed, voluminous (data from various periods of time) and lightly aggregated or not

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